

Lab Goal : This lab was designed to teach you more about graphs, mazes, and recursion.

Lab Description : You have found the StrongHold! Now you must navigate this StrongHold to find the ender portal to kill the Ender Dragon. Find the shortest path through the “maze” from your position to the portal. You can only move in the 4 cardinal directions. There is also no guarantee that the StrongHold you are in will have a portal at all.

Input : The first line will contain a single integer n that indicates the number of data sets that follow. Each data set will start with a 2 integers, r and c , representing the number of rows and columns in the following maze.

The next r lines will contain c characters, representing the StrongHold you are to navigate. It will be made up of the following characters:

'#'-Wall	'K'-Skeleton	'L'-Slime	'W'-Cobweb
'S'-Spider	'E'-Enderman	'C'-Cave Spider	
'Z'-Zombie	'P'-Portal	'F'-Silver Fish	
'.'-Empty Space	'T'-Steve (You)	'R'-Creeper	

You can travel only through empty spaces, you begin on Steve, and must end on the portal. You cannot travel within 3 blocks of skeletons or creepers, within 1 block of any other monster, or through walls or cobwebs.

Output : You will output the minimum number of steps to the portal, and if there is no safe path to the portal output "You can't afford, ford, ford, my diamond sword, sword, sword"

Example Input File

```

8
5 6
#####
#T.E.#
#.W..#
#...L#
##P###
10 10
#####
#...T....K
#LL..WW..#
#LL#..W..#
R...#...E#
#.C.....#
#..##...##
K.....K
#.EE.P..F#
#####
3 1
T
#

```

```

P
10 10
#####
#...T....K
#LL..WW..#
#LLZ..W..#
R...#...E#
#.C.....#
#..##...##
K.....K
#.EE.P..F#
#####
5 5
T.R..
P....
.....
.....
...E.
21 21
..#####
..#.#.W.....#..R#
#.#.#.#####.#
#...#.....#...#.#.#
#.#.#.#P###.###.#.#
#K#...#...#.#.....#
#.#####.###.#.#####
#.#.#.....#...Z.#
#.#.#####.#.###.#.#
#.F.#...#...#...#.#.#
#.#####.#.#####E#.#
#...#...#.#.....#
#.F###.##...###.#.#####
#F#L..L.....#.....#
#F#.#####.....W###.#
#.#...#...#...#.#...#
#C#.#.#.###.###.###.#
#...#.#.#.....#.....#
#.#.#S#.#.#.###.#.###
#.#.K...#.#...#.#...T
#####.#
5 5
T....
.....
.....
.....
.....
8 8
T...K...
.....
.....
....R...
.....

```

.....
.....
.....P

Example Output to Screen

```
4
8
You can't afford, ford, ford, my diamond sword, sword, sword
You can't afford, ford, ford, my diamond sword, sword, sword
You can't afford, ford, ford, my diamond sword, sword, sword
32
You can't afford, ford, ford, my diamond sword, sword, sword
14
```